

A TOPOS FOR ÉTALE-FINITE HEYTING ALGEBRAS

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Lattices of truth-values in toposes

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3. The effective topos Eff .

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General problem: What are the possible lattices which arise in this way?

All lattices of truth-values are *Heyting algebras*:

Definition

An algebra $(H, \wedge, \vee, \rightarrow, 0, 1)$ is called a **Heyting algebra** if:

1. $(H, \wedge, \vee, 0, 1)$ is a (distributive) lattice.
2. The following law holds for all $a, b, c \in H$:

$$a \wedge c \leq b \iff c \leq a \rightarrow b.$$

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Complete Heyting algebras are called *frames* or *locales*: for example

1. $\mathcal{O}(X)$ – opens of a topological space X ;
2. $\text{Con}(A)$ – congruence lattices for congruence-distributive algebras.

Which Heyting algebras

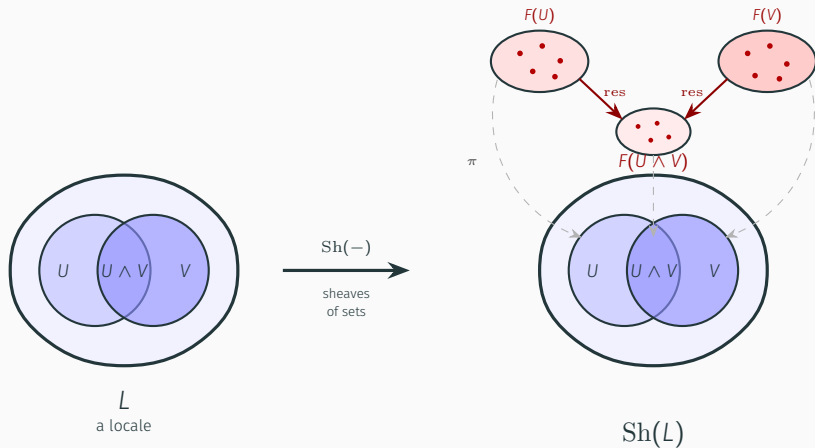
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$$\mathrm{Sub}_{\mathrm{Sh}(L)}(1) \cong L.$$

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Logically: interpreting a higher order theory in its propositional fragment – analogous to **higher-order uniform interpolation**.

Topos-theoretically: is there a **good quotient of free toposes** which produces the free Heyting algebras $\mathcal{F}(X)$?

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This was the approach suggested by D. Patariaia. But eliminating higher-order sentences is much more difficult: equality types, extensionality, and various other headaches.

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(Please let us know if you do!)

Étale-finite Heyting algebras

We work with a class introduced by E. Kuznetsov (2024):

Definition (Étale-finite Heyting algebra)

A Heyting algebra H is **étale-finite** if there is a finite Heyting algebra H_0 and a Heyting algebra homomorphism $f: H_0 \rightarrow H$ satisfying **Jibladze's law**: for every $x \in H$:

$$\bigvee_{x_0 \in H_0} (x \leftrightarrow f(x_0)) = 1.$$

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Conceptually: there are **finitely many definable truth-values**.

Example

Finite algebras are étale-finite. For subdirectly irreducible, étale-finiteness is just finiteness.

We also work with their **duals**:

Definition (Esakia space)

An ordered space $(X, \leq, \tau)$ is an **Esakia space** if it is (1) compact, (2) Priestley separation and (3) $\downarrow W$ is clopen whenever W is clopen.

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$$f(x) \leq y \implies \exists !x' \geq x, f(x') = y;$$

equivalently $\uparrow x \cong_f \uparrow f(x)$.

For example:

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We work with maps which are local homeomorphisms both for the order and the topology:

Definition (Spectral local homeomorphism)

A map $f: (X, \leq, \tau) \rightarrow (Y, \leq, \tau)$ between Priestley spaces is a **spectral local homeomorphism** if f is a local homeomorphism for the spectral topology.

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Proposition (Abbadini, A., Arrieta)

f is a spectral local homeomorphism if and only if it is a local homeomorphism for the Priestley/patch topology, and a strict p -morphism.

Building toposes

Our analogue of $\mathbf{Top}_{\text{LH}}$ – topological spaces with local homeomorphisms – is then

$$\mathbf{Esa}_{\text{SLH}}$$

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Not such a nice category: lacks terminal objects and equalizers. But its **slices** are good.

For X an Esakia space:

$$\mathbf{Esa}\acute{\mathbf{E}}\mathbf{t}(X) = \mathbf{Esa}_{\text{SLH}}/X$$

Starters - finite limits

Because our morphisms are nice, the limits in this category behave much better than in general for Esakia spaces:

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Theorem (Abbadini, A., Arrieta)

For every Esakia space X , the category $\mathbf{Esa\acute{E}t}(X)$ is finitely complete:

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2. (Binary product) *If $f: Y \rightarrow X$ and $g: Z \rightarrow X$ are two objects in $\mathbf{Esa\acute{E}t}(X)$, then the pullback $Y \times_X Z$ as ordered spaces with the two projection maps is the product in $\mathbf{Esa\acute{E}t}(X)$.*

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3. (Equaliser) Whenever $f: Y \rightarrow X$ and $g: Z \rightarrow X$ are two objects of $\mathbf{Esa\acute{E}t}(X)$, and $p_1, p_2: Y \rightarrow Z$ are two spectral local homeomorphisms making the two triangles in the diagram commute,

$$\begin{array}{ccc} Y & \begin{array}{c} \xrightarrow{p_1} \\ \xrightarrow{p_2} \end{array} & Z \\ & \begin{array}{c} \searrow f \\ \swarrow g \end{array} & \\ & X & \end{array}$$

then

$$\text{Eq}(p_1, p_2) = \{y \in Y : p_1(y) = p_2(y)\},$$

together with the inclusion into Y , is the equaliser of p_1 and p_2 in $\mathbf{Esa\acute{E}t}(X)$.

Any inclusion of a **clopen upset** $U \hookrightarrow X$ is a spectral local homeomorphism, and these are all the injective ones. So we have:

Theorem (Abbadini, A., Arrieta)

Given an Esakia space, $\mathcal{E} = \mathbf{Esa}\acute{\mathbf{E}}\mathbf{t}(X)$, we have:

$$\mathbf{Sub}_{\mathcal{E}}(1) \cong \mathbf{ClopUp}(X).$$

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Via Esakia duality, this gives us **every Heyting algebra**. Incidentally, $\mathbf{Esa\acute{E}t}(X)$ is always a **Heyting pretopos**.

The main dish - power objects

What about power objects?

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More difficult to construct. Basic idea: use a **Vietoris-like** construction (typical in the theory of (co)free objects in theories such as Heyting algebras and modal algebras).

Given $f: Y \rightarrow X$ an object in $\mathbf{Esa\acute{E}t}(X)$, where X is étale-finite:

$$P_f(Y) := \{(x, K) : x \in X, K \text{ an upset of } Y \text{ with } K \subseteq f^{-1}[\uparrow x]\}.$$

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we define the partial order $\preceq$ by setting

$$(x, K) \preceq (x', K') \iff x \leq x' \text{ and } K' = K \cap f^{-1}[\uparrow x'].$$

and let it inherit the topology from the product $X \times \mathcal{V}(Y)$ – the Vietoris space of Y .

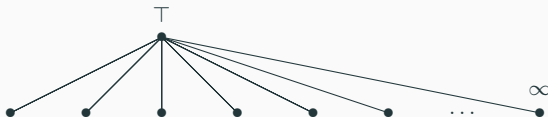
Power objects – Example

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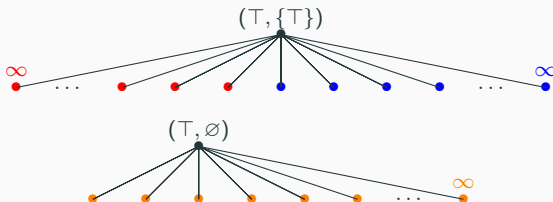


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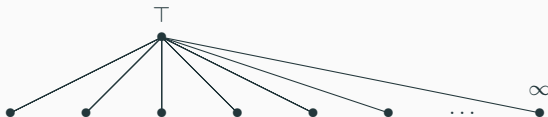


Then $P_{1_X}(X)$ will be:

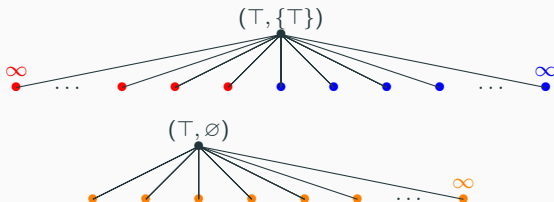


Power objects – Example

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Then $P_{1_X}(X)$ will be:



Blue points – $(x, \{T\})$;

Red points – $(x, \{x, T\})$.

Orange points – $(x, \emptyset)$.

Theorem (Abbadini, A., Arrieta)

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The key difficulty lies in showing that $P_f(Y)$ is actually a **compact space**. But with that we obtain:

Theorem (Elementary topos for étale-finite)

For every étale-finite Heyting algebra H , with $X = \text{Spec}(H)$, $\mathcal{E} = \mathbf{Esa\acute{E}t}(X)$ is an elementary topos, such that $\text{Sub}_{\mathcal{E}}(1) \cong H$.

Finitely propositional toposes

Two interrelated questions:

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2. Can we get any more toposes this way?

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A category (with a terminal object) is **(finitely) propositional** if every object Y admits a jointly epimorphic (finite) family of morphisms from subterminal objects

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Theorem (Abbadini, A., Arrieta)

For every Esakia space X , the category $\mathbf{EsaEt}(X)$ is finitely propositional.

Through studying the structure of the lattice of truth-values, we get (Abbadini, A., Almeida):

Theorem (Finitely propositional $\leftrightarrow$ étale-finite)

For a Heyting algebra H , the following are equivalent:

1. $\mathbf{EsaÉt}(\mathrm{Spec}(H))$ is a topos;
2. H is isomorphic to the lattice of truth values of some finitely propositional topos;
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This hints at a possible (2-)equivalence of categories, which we have not yet investigated.

Final remarks

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If there is a counterexample, it should morally be the free Heyting algebra $\mathcal{F}(\omega)$, so it should fight us on one of these fronts.

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1. **Arithmetic and type theory**: What happens to these questions if we demand the presence of a natural numbers objects?
2. **Higher topos theory**: What happens if one replaces lattices of truth-values by a “large” object, like an elementary topos in a 2-topos?

Thank you!



Questions?